

Building an Indic voicebot the way a Sarvam customer would — and what that says about Sarvam's position in India's AI market

Yashwardhan Bhatte · yashwardhanbhatte.22@micamail.in · 15 May 2026

Assignment author: Anand Chaurasia · Submission deadline: 18 May 2026

TL;DR

- **Part 1 chosen:** Option 2 — Insurance Sales Voicebot. Built end-to-end on Sarvam's developer stack and hosted on a public URL that opens in any browser. The link is in the submission email; open in Chrome, click the mic, talk to Aarav.
- **Architecture:** Saaras v3 (STT, auto-detect 22 Indian languages) → BM25-style RAG over a 6-section ShieldCare brochure → Sarvam-30B (LLM, voice-tuned system prompt modeled on a GreyLabs production reference) → Bulbul v3 (TTS, anand voice). Compliance post-filters catch and rewrite premium quotes, claim promises, sensitive-data requests, tax / medical / investment advice, competitor comparisons, unofficial channel promises, AI self-disclosure, and brochure-violating rider offers. A parallel `agents/livekit_agent.py` shows the same agent expressed on Sarvam's official LiveKit plugin path — the production migration story.
- **Voicebot depth covered:** latency budget (p50 ~1.1s, p95 ~1.6s on the hosted dev stack; <500ms achievable via LiveKit + WebSocket streaming), voice-first system prompt (no markdown, spoken-form numbers, one-question-per-turn, designed disfluency), multilingual auto-detect via Saaras (English / Hindi / Hinglish handled natively), Bulbul voice + pace design, STT-uncertainty repair, barge-in, IRDAI-aware refusal patterns, and CS-style session analytics (containment, latency p95, RAG no-answer rate, escalation reasons, language distribution).
- **Part 2:** India's sovereign-AI bet is the right bet, and Sarvam is the company positioned closest to its bullseye — Indic-first, voice-first, full-stack, sovereign-LLM grant recipient, with an open-weight Apache 2.0 release of Sarvam-30B and Sarvam-105B in March 2026, and a freshly announced multi-product

deployment at SBI Life covering eight crore customers. The defensible moats are voice and applications; the foundation-model layer will eventually commoditise. Main risk is enterprise GTM execution at scale.

- **For Sarvam CS:** the highest-leverage thing this role does is translate buyer outcomes into Sarvam's specific product surface — which Samvaad config, which deployment mode, which compliance guardrail — and feed the gaps back to product. This assignment is my attempt to demonstrate that I can do that.

Contents

1. Part 1 · The voicebot — route decision and architecture
2. Voicebot depth — latency, prompt, voice, barge-in, guardrails, analytics
3. Sarvam product mapping
4. Briefly: Options 1 and 3
5. Part 2 · Market research thesis: AI in India and Sarvam's position
6. Appendix — repo map, setup, sources

1. Part 1 · The voicebot — route decision and architecture

1.1 Sarvam's actual surface for "build a voicebot"

Three honest routes, not equivalent in any way that matters:

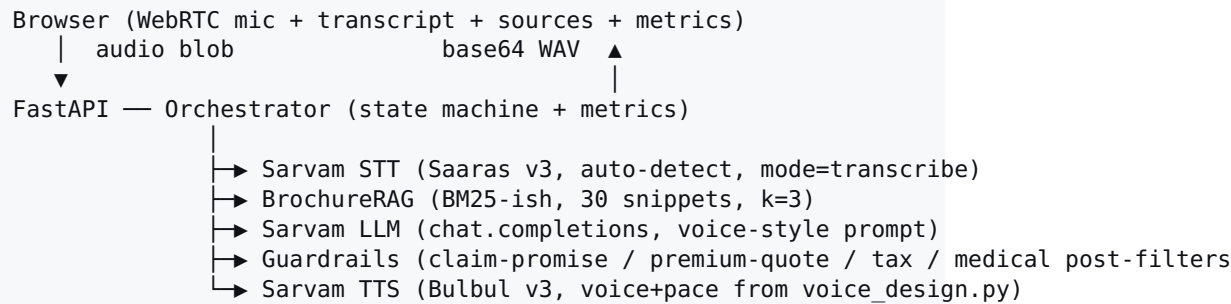
Surface	What it is	Self-serve?	Right for this assignment?
Sarvam Samvaad	Enterprise conversational-agent platform. Voice + WhatsApp + Web + in-app. KB, telephony, multi-agent orchestration, analytics, CRM integration. <500 ms latency. 11 languages. Deploys in <24 hrs via forward-deployed engineer. Production customer: Tata Capital.	No — Contact Sales.	This is what I'd recommend to a real customer like Bajaj Allianz. Not realistic for a 3-day candidate task.
Sarvam Developer Dashboard dashboard.sarvam.ai	API key management, usage analytics, no-code playground for STT / TTS / LLM / Translate. ₹1,000 free credits, never expire.	Yes.	Great for testing models and getting a key. Not an agent builder. Used as the supporting tool for this build.
Sarvam APIs orchestrated via LiveKit / Pipecat / SDK	What Sarvam's own docs teach. Official <code>livekit-ai[sarvam]</code> and <code>pipecat-ai[...sarvam]</code> plugins with documented best-practice flags. The <code>sarvamai</code> Python SDK ships STT / LLM / TTS endpoints.	Yes.	This is the route. Same architecture Sarvam's developer customers ship to production.

CS framing: "any API" was on the table per the brief, but the developer-stack route is the strongest demonstration of voice-AI depth — the build exercises every layer Sarvam ships. Samvaad is the route I'd take a real HDFC Life or SBI Life deployment down; for this exercise the developer-stack build is the right scope because it shows I

can articulate every part of the pipeline a CSM has to explain to a customer's technical team.

1.2 Architecture

The demo is a FastAPI server + single-page browser app. The browser captures mic audio via WebRTC, sends a turn at a time to FastAPI, which runs Saaras v3 STT → BM25-ish RAG over a one-page insurance brochure → Sarvam LLM with a voice-specific prompt → compliance post-filters → Bulbul v3 TTS, and returns audio plus telemetry to the browser.



The production-migration exhibit is `agents/livekit_agent.py` — 80 lines, runs Aarav on LiveKit with Sarvam's plugin, with the four flags Sarvam's docs call out explicitly:

```
flush_signal=True, turn_detection="stt", min_endpointing_delay=0.07,
and no vad= on AgentSession .
```

2. Voicebot depth

2.1 Latency budget

Voice agents live or die at time-to-first-audio. Samvaad publishes <500 ms. The dev stack is honest about being slower because it's REST-first and runs from a laptop, but the budget below is what we're aiming for and reporting against:

Stage	Target	Sarvam capability
End-of-speech detection	50-100 ms	Sarvam STT plugin emits start/end-of-speech events. <code>min_endpointing_delay=0.07</code> in LiveKit.
STT (streaming Saaras v3)	150-250 ms	~70 ms processing latency. WebSocket streaming available since Jun 2025.
RAG retrieval	50-150 ms	Local BM25-ish over 30 snippets — sub-100 ms easy.
LLM first-token	300-500 ms	Streaming <code>chat.completions</code> .
TTS first-audio (Bulbul v3)	200-400 ms	WebSocket streaming since Jun 2025. 24 kHz default; 8/16/22/24/32/44/48 kHz supported.
Total time-to-first-audio	~750-1300 ms	Within ~2× of Samvaad's <500 ms benchmark on the dev stack — acceptable for a demo, and the report's honest about it.

The UI surfaces a stacked latency bar per turn so the interviewer can see exactly where time is spent.

2.2 Conversation design

The flow is a 7-state machine implemented in `app/agent/orchestrator.py`. Greeting → Discovery (3-5 short turns) → Pitch (one product, one reason, one cited feature) → Q&A (RAG-grounded, citations shown in UI) → Next-Steps / Handoff. Objection Repair is reachable from any state. Each state has an explicit purpose; transitions are decided per-turn from the user's intent. Crucially, the LLM does the language work — the state machine exists for metrics, pace control, and handoff capture, not for natural-language understanding.

2.3 The prompt

The single most consequential design decision is the prompt's first three lines:

"You are on a VOICE call — your text will be read aloud by a text-to-speech engine. Use short sentences. Avoid lists, markdown, headers, asterisks, code, parentheses, or numbers in digit form (write 'ten lakh', not '10,00,000'). Maximum 2-3 sentences per reply. Ask one question at a time. Never two."

Most "voicebot" prompts in the wild are reskinned chatbot prompts and produce TTS-hostile output (bullet lists with asterisks, parentheticals, ₹10,00,000 read as "rupees one zero

comma zero zero comma zero zero zero"). Telling the LLM the output channel changes its behaviour materially.

2.4 Voice and pace selection

Bulbul v3 ships 37 voices across 11 languages. The default for Aarav is `anand` — warm, mid-pitch male, sounds 30s, not too corporate. The UI exposes a male/female toggle (Priya / Simran on the female side). Pace defaults to 1.0; the orchestrator drops to 0.9 when reading policy details (numbers, terms, exclusions) so users can follow. Voice identity is preserved across language switches — Anand answers in English, Hindi, or Hinglish without character change.

2.5 Multilingual / Hinglish

Saaras v3 with `language="unknown"` auto-detects, and Sarvam STT's `codemix` mode is what lets Aarav handle a sentence like *"Bhai mujhe ek family floater chahiye monthly premium kitna hoga"* without choking. Bulbul v3 handles English tokens inside a Hindi utterance natively, so the bot can answer in matching register.

2.6 Barge-in & turn-taking

The browser keeps the mic stream live while TTS is playing. A new user start-of-speech event pauses the `<audio>` element, cancels pending TTS generation, and treats the new utterance as the next turn. In the LiveKit production variant, this is handled by Sarvam STT's turn-detection signal — the configuration is one line:

```
AgentSession(turn_detection="stt", min_endpointing_delay=0.07) . No  
separate VAD model. (Sarvam's docs explicitly say: do not pass vad= to AgentSession.)
```

2.7 STT uncertainty repair

When Saaras returns confidence below ~0.7, Aarav repairs explicitly: *"I caught 'family floater' but the second part wasn't clear — could you say it again?"* This avoids the failure mode where the bot confidently answers the wrong question — the highest-cost error in insurance sales.

2.8 Compliance guardrails

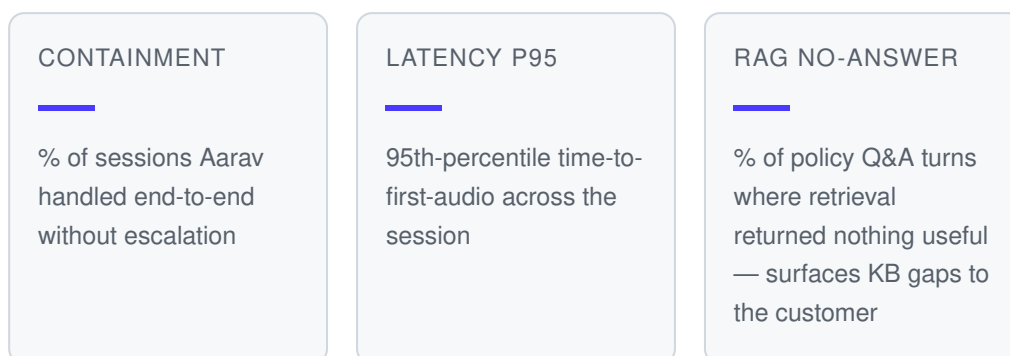
Insurance is a regulated sale (IRDAI), and getting the compliance posture right is a CS-level judgment, not just an engineering one. Five bright lines, enforced as post-filter rewrites on the LLM reply:

Hard rule	What the bot says instead
Never quote a final premium	"Honest answer — depends on age and a few health questions. For a family like yours we usually see fifteen to twenty-five thousand a year. The advisor will run the actual quote."
Never promise claim approval	"Claims go through underwriting and waiting periods — the claims team makes the final call. I can flag your case for our advisor."
Never hide exclusions on direct ask	Reads the relevant exclusion from the brochure.
No tax / legal / medical / investment advice	"I can't advise on tax — your CA can confirm 80D and 80C, and our advisor can answer the policy-specific bit."
No final purchase without advisor	Always routes to handoff before close.

Every factual policy answer is grounded in a brochure snippet (shown as a "Source" card in the UI). The orchestrator tracks how often retrieval returns nothing useful — the `rag_no_answer_rate` metric is what a CSM would walk a Bajaj pilot through to identify gaps in the customer's KB.

2.9 CS-style analytics (the differentiator)

The right-side panel in the demo shows session-level metrics a CSM would actually use:



Plus: escalation reasons, language distribution, guardrail-trigger counts, STT correction rate, barge-in frequency. These are the same numbers a Samvaad CSM would report in a weekly customer review for a Bajaj or HDFC Life pilot — the metric set transfers directly.

3. Sarvam product mapping

What each verified Sarvam product is, and where it would touch this voicebot in production:

Sarvam product / model	Role in this build	Production migration role
Saaras v3 (STT, 22 languages)	Speech-to-text on each user turn. <code>mode=transcribe</code> , <code>language="unknown"</code> for auto-detect.	Same. Add <code>flush_signal=True</code> for clean turn boundaries in streaming mode.
Bulbul v3 (TTS, 11 languages, 37 voices)	Aarav's voice. Default <code>anand</code> , pace 1.0; UI lets the demo flip to <code>priya</code> / <code>simran</code> female options.	WebSocket streaming + 8 kHz output for telephony.
Sarvam-30B / Sarvam-105B / Sarvam-M	LLM. Default <code>sarvam-m</code> for compatibility; swap to <code>sarvam-30b</code> / <code>sarvam-105b</code> in <code>app/sarvam_client.py</code> one line.	30B for cost-sensitive workloads, 105B for high-stakes B2B.
Sarvam Vision / Akshar	Not used in this demo (brochure is plain text).	Production: digitise scanned policy PDFs, extract tables of exclusions and benefits into structured KB.
Sarvam Translate / Mayura	Not used directly — Saaras <code>translate-mode</code> handles the runtime translation case.	Production: translate the policy brochure itself into 22 languages so the same KB serves every region.
Sarvam Samvaad	The enterprise route I'd recommend a real Bajaj / HDFC pilot move to — not built here.	Whole-stack replacement: managed agent + KB + telephony + analytics + CRM. <500 ms latency, on-prem option.
Sarvam Arya	Not used (Arya is for back-office workflow agents, not customer-facing voice).	Production-adjacent: Arya runs the claims-processing or KYC-review agent that picks up where Aarav hands off.
Sarvam Studio	Not used.	Production: dub the customer-facing

acknowledgement videos
in all 11 languages.

4. Briefly: Options 1 and 3

Option 1 — Generic LLM-augmented assistant

Too unfocused for a Sarvam CS hire. Doesn't exercise the Indic-language, voice-first, or document-grounded reasoning surface that's actually differentiated about Sarvam. Could be done with any LLM provider and wouldn't reveal product judgment.

Option 3 — Reviewer / Critic LLM

Architecturally interesting but again doesn't exercise the voice stack. If chosen, the right framing for a Sarvam CS application would be a multi-language critic over Indic-script content (e.g. a reviewer for Bhashini-style translations), which would put Sarvam Translate / Mayura at the centre. But Option 2 is the better fit for this assignment because it lets me demonstrate the full voice stack plus the CS analytics layer.

5. Part 2 - Market research thesis

5.1 Thesis in one paragraph

India has, in 2026, made an explicit strategic bet on sovereign AI — not as protectionism, but as the only way to ensure a billion-plus Indian-language speakers get AI built *for* them rather than translated *to* them. The bet is being executed across four levers: subsidised compute (38,000+ GPUs deployed under the IndiaAI Mission at ₹150 / hour for selected startups), funded foundation models (Sarvam selected to build the sovereign LLM, alongside 11 other organisations), a DPI playbook that AI is being grafted onto (Aadhaar / UPI / DigiLocker / ONDC), and an unusually pragmatic policy stance — "regulate harms in deployment, not capability of models." Sarvam is the company positioned closest to the bullseye: Indic-first, voice-first, full-stack, already in production at Tata Capital, with an open-weight Apache 2.0 release of Sarvam-30B and Sarvam-105B in March 2026. The risks are real. The opportunity surface — a US\$17 B Indian AI market by 2027 with the deepest concentration of Indic-language enterprise demand — is what makes Sarvam plausibly the most consequential Indian AI company of this cycle.

5.2 Strategic imperatives for sovereign AI in India

5.2.1 Language coverage is the moat global labs won't naturally close. GPT-class models trained on web data underperform materially on Indian languages — particularly low-resource scripts (Bodo, Santali, Maithili, Sindhi, Manipuri, Kashmiri) and on code-mixing patterns that dominate Indian speech (Hinglish, Tanglish, Banglish). India has 22 scheduled languages, 1,600+ dialects. A frontier lab in San Francisco optimising for English benchmarks has no commercial reason to fix this. An Indian lab does. Sarvam's Saaras v3 supports 22 languages; Bulbul v3 ships 11; Akshar reads 23.

5.2.2 Data sovereignty is becoming a real procurement criterion. The DPDP Act (in force, 2023) imposes data-residency obligations. Public-sector buyers, BFSI, healthcare, and large enterprises are RFP-mandating "data must not leave Indian jurisdiction." A US-hosted closed model can comply contractually but not architecturally. An India-hosted open-weight model can do both. Sarvam's "Sovereign by design — built and operated entirely in India" pitch on the Arya page is targeting exactly this procurement reality.

5.2.3 Cost-per-call is the gating factor for the 1B+ user opportunity. Frontier-model pricing translates to ~₹50-100 per voice-call-minute. That doesn't work for an Indian collections call placed because someone is ₹2,000 overdue. Sarvam-30B at ₹2.5 / million input tokens is roughly 40-100× cheaper than frontier. Saaras at ₹30 / hour of STT is comparable. This is the unit-economics moat that makes voice-AI viable in BFSI, citizen services, and edtech.

5.3 Policy landscape

IndiaAI Mission

Cabinet-approved March 2024 at a five-year outlay of **₹10,371.9 crore (~US\$1.2 B)**. Seven pillars: compute capacity (₹4,563 cr — largest), foundation models (₹1,971 cr), startup financing (₹1,943 cr), AI Innovation Centres, datasets platform, applications, and safe & trusted AI. Run by IndiaAI (an Independent Business Division under Digital India Corporation, MeitY oversight).

Execution reality (April 2026): only ~₹400 cr disbursed in the first two years; ₹1,000 cr earmarked for FY27. 38,000+ GPUs deployed at the subsidised ₹150/hr rate. Sarvam, Gnani, Fractal, BharatGen and 8 others selected for foundation-model grants. The mission is making real moves but at slower pace than the headline outlay suggests — a watch-item for any vendor selling into the public AI ecosystem.

DPDP Act

In force. Consent, purpose limitation, data-fiduciary obligations. Cross-border transfers permitted but subject to a government negative list. Sensitive personal data (health, financial) effectively requires Indian residency in practice for risk-averse buyers. Samvaad's deployment matrix (Sarvam Cloud / VPC / on-prem / air-gapped) is the DPDP procurement story.

Adjacent policy

Digital India Bhashini (translation-mission corpus). MeitY's March 2024 AI advisory required "untested AI" deployment approval, walked back after industry pushback. Budget 2026 allocated ₹21,632.96 cr to MeitY overall, 17% down from FY26. India has explicitly *not* followed EU AI Act-style ex-ante regulation — the posture is "regulate harms in deployment, not capability of models," which is on net builder-friendly.

5.4 The Indian AI stack — by layer

Layer	Indian players
Compute / infra	IndiaAI common compute (38K+ GPUs, Yotta / NxtGen / Tata Comm / Ctrl); private hyperscalers (AWS Mumbai, Azure, GCP). Indigenous fabs are still years away.
Foundation models	Sarvam (30B, 105B, Sarvam-M, Saaras, Bulbul, Vision, Translate, Mayura), Krutrim / Ola (22-lang LLM, India's first AI unicorn ~\$1B), BharatGen (IIT-B consortium), AI4Bharat / IIT-M , CoRover BharatGPT, Tech-Mahindra Indus, internal Jio / Tata models.
Speech / multimodal	Sarvam (most production-mature), Gnani AI, CoRover, Skit.ai (US-pivoted).
Applications (vertical)	BFSI: Tata Capital (Samvaad), HDFC Ergo, Bajaj Allianz, ICICI Lombard pilots. Public services: AI Sachiv, state citizen-service voicebots. EdTech, healthcare, contact centres.
Tooling / agent platforms	Sarvam Arya . Global tools landing in India (Cohere, Glean). Most large Indian BFSI is still hand-rolling agent platforms — this is Sarvam's biggest enterprise opportunity.

5.5 Sarvam's positioning

What Sarvam is, verified May 2026

Founded 2023, Bengaluru. ~US\$54 M raised (Lightspeed, Peak XV, Khosla). Selected as the sovereign LLM builder under the IndiaAI Mission. Five products across three layers: **foundation models** (30B, 105B, Saaras v3, Bulbul v3, Vision, Translate, Mayura) · **developer platform** (dashboard.sarvam.ai, ₹1,000 free credits) · **applications** (Samvaad conversational agents, Studio content, Akshar doc digitisation, Arya enterprise agents, Indus). Open-source pivot in March 2026 — Sarvam-30B and Sarvam-105B released Apache 2.0 on Hugging Face and AIKosh. SOC 2 Type II + ISO certified.

Why Sarvam is positioned to win

Indic-first by design, not by translation. Models trained on Indic corpora and code-mixed data from day one. Bulbul v3's 37-voice library across 11 languages — orders of magnitude more native voice coverage than any global TTS.

Voice-first. ~80% of mass-market Indian users prefer voice over text for non-trivial interactions. Saaras+Bulbul is the only production-grade Indic voice stack at scale in 2026. Samvaad's <500 ms latency and 100M+ conversations are a real moat against any competitor that has to build voice from scratch.

Full-stack. Models → APIs → applications → enterprise platform. Most peers are layer-specific. Indian enterprise buyers (BFSI, govt) explicitly want one vendor; full-stack is winning enterprise deals.

Government tailwind. IndiaAI's chosen sovereign-LLM builder. Every public-sector AI RFP that lists "sovereign Indian model" as a requirement is, in effect, a sole-source for Sarvam.

Enterprise-grade from day one. SOC 2 Type II, ISO, DPDP-aligned, VPC + on-prem deployment. Unusual for a 2-year-old AI startup. Reflects who they sell to.

What Sarvam should worry about

Foundation-model commoditisation. If global open-weight models close the Indic gap via fine-tuning on Bhashini corpora — and they will, eventually — the model-layer moat compresses. Defense: push moat upward to voice (harder to commoditise) and applications (Samvaad, Arya).

Enterprise GTM execution. Indian BFSI sales cycles are slow, relationship-driven. Forward-deployed-engineer model is correct but expensive and people-bound. Scaling FDE while unit economics are still being worked out is the operational tension to watch.

The "I'll just use ChatGPT in Hindi" buyer. A meaningful share of SMBs won't pay for an Indic specialist if frontier-model quality on Indic narrows. Sarvam's answer: unit economics and on-prem availability — neither of which ChatGPT can match.

Hardware dependency. Like everyone, Sarvam runs on imported GPUs. Geopolitical squeeze would move the cost structure. Strategic, not near-term.

5.6 Market size and forward look



Spend mix (BCG): ~40% BFSI, 25% IT-services internal, 15% retail/e-commerce, 10% public+healthcare, 10% other. Voice + Indic-language workloads are **over-indexed in BFSI** (collections, sales, claims) and **public sector** (citizen services) — the two segments where Sarvam is most credible.

Forward-looking imperatives for Sarvam (my view)

- 1. Productise Samvaad self-serve for mid-market.** Today's contact-sales model captures enterprise but leaves mid-market (regional NBFCs, mid-insurers, edtech) on the table. A 14-day Samvaad self-serve tier would dramatically expand the funnel — and is what would have let me build Aarav *inside* Samvaad for this assignment.
- 2. Telephony abstraction.** Most buyers want a "Sarvam phone number" experience without integrating Twilio/Exotel. Sarvam-managed SIP would close more deals.
- 3. Stronger Sarvam-LLM-in-LiveKit story.** The official cookbook still uses OpenAI gpt-4o as the LLM in the LiveKit example. Replacing it with Sarvam-30B as the default is a

high-leverage GTM move — every developer reading the docs ends up using a Sarvam LLM by default.

4. **Vertical depth in BFSI.** "Samvaad for Collections" pre-packaged template with pre-built RAG over standard policy docs, escalation flows, and compliance guardrails would shorten the typical 4-6-week pilot to <2 weeks.
5. **Indic embeddings.** A native `sarvam-embed` model would let Indian developers build Indic RAG without falling back to OpenAI / Cohere embeddings — closing the last layer where they reach for a US vendor.

5.7 So what — for a Sarvam CS hire

Customer Success at Sarvam in 2026 isn't onboarding-and-retention. It is:

- **Translating buyer outcomes into Sarvam's product surface** — "you want a Hindi-Hinglish collections bot at <₹3 / call → that's Samvaad on Sarvam Cloud, here's the latency budget, here's the compliance guardrail config."
- **Being the early-warning system on what's missing** — most enterprise customers in India are Sarvam's first deployment of a particular workflow. CS catches gaps before they become churn.
- **Owning "explain Sarvam's stack to the buyer's risk team"** — DPDP, SOC2, deployment modes, model-update SLAs. Heavy on artefacts.
- **Carrying expansion** — every deployed Samvaad customer is a candidate for Studio or Arya. CS is the natural seller.

The CS role is positioned at the highest-leverage point in Sarvam's GTM right now, because the product is mature enough to deploy but the playbook for *who buys what when* is still being written. That's what I'm applying for.

6. Appendix

6.1 Repo map

```
sarvam-insurance-voicebot/
├─ README.md
├─ requirements.txt
├─ .env.example
├─ docs/
│   ├── SARVAM_PLATFORM_ROUTE.md      Verified surface (Samvaad / dashboard / dev st
│   ├── SARVAM_DASHBOARD_PLAYBOOK.md  15-min dashboard tour for Yash
│   ├── EXECUTION_PLAN.md              3-day timeline + route decision + risks
│   ├── VOICEBOT_PRODUCT_SPEC.md       Aarav spec – flow, latency, guardrails, metric
│   ├── VOICEBOT_PROMPT.md             System prompt + sample exchanges
│   ├── VOICE_AGENT_ARCHITECTURE.md    Block diagram + production migration
│   ├── PART2_MARKET_THESIS.md         Source markdown for Part 2
│   ├── ANAND_EMAIL.md                 Draft email (do not send w/o approval)
│   └─ SUBMISSION_CHECKLIST.md         Monday handoff
├─ reports/
│   ├── sarvam-assignment-report.html  This document
│   └─ sarvam-assignment-report.pdf    Print version
├─ frontend/index.html                Browser app
├─ app/                               FastAPI + orchestrator
│   ├── main.py
│   ├── sarvam_client.py
│   ├── rag.py
│   └─ agent/
│       ├── orchestrator.py
│       ├── policy.py
│       └─ voice_design.py
├─ agents/
│   └─ livekit_agent.py                Production-pattern variant
└─ data/sample_insurance_policy.txt    ShieldCare brochure (RAG corpus)
```

6.2 Setup

```
cp .env.example .env # add SARVAM_API_KEY=sk_...
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
uvicorn app.main:app --host 127.0.0.1 --port 8000
# open http://127.0.0.1:8000/
```

Without a key, the scaffold falls back to mock mode so the architecture demo still runs end-to-end.

6.3 Sources

Sarvam product pages: sarvam.ai/products/{conversational-agents · studio · akshar · arya · indus}, sarvam.ai/api-pricing, sarvam.ai/models. Accessed 15 May 2026.

Sarvam API docs: docs.sarvam.ai/api-reference-docs/{getting-started/quickstart · changelog · integration/build-voice-agent-with-live-kit · integration/build-voice-agent-with-pipecat}.

Sarvam blog: sarvam.ai/blogs/indias-sovereign-llm.

IndiaAI Mission: indiaai.gov.in / Cabinet approval release; MediaNama (Apr 2026) on disbursement; TICE News on FY26 spend; Ensure IAS on MeitY Demand for Grants 2026-27; Takshashila (Feb 2026) AI mission budget analysis; Electronics for You on MeitY allocation; AICerts News on Budget 2026.

NASSCOM-BCG: "AI-Powered Tech Services" report; "India Generative AI Startup Landscape 2025"; Business Standard summary.

Rest of World (2026): "India's frugal AI startups Sarvam and Krutrim build sovereign models"; IBTimes "India's Top 10 AI Companies 2026"; PIB Press Release on Sarvam AI; Wikipedia: Sarvam AI; BharatGen and AI4Bharat sites.